

$r_f = 0.55 \text{ cm}$, $LHR = 300 \frac{\text{W}}{\text{cm}}$, $K_f = 0.025 \frac{\text{W}}{\text{cm-K}}$, $E = 200 \text{ GPa}$

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$\nu = 0.35$
 $\alpha = 10 \text{E-6 K}^{-1}$

a) σ_{max} ?

b) $\sigma_f = 175 \text{ MPa}$, how far do cracks extend? $\eta = \frac{r}{0.55}$

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$\sigma^* = \frac{10 \text{E-6} \cdot 200 (\Delta T_F)}{4(1-0.35)}$ ✓

~~$\Delta T_F = \frac{LHR}{4\pi K_f} = 954.93 \text{ K}$~~

$\Delta T_F = \frac{LHR}{4\pi K_f} = 954.93 \text{ K}$ ✓

$\sigma^* = 0.735 \text{ GPa} = 735 \text{ MPa}$ ✓

$\sigma_r = -\sigma^* (1-\eta^2) = -\sigma^* (1-(\frac{r}{0.55})^2)$

$\sigma_\theta = -\sigma^* (1-3\eta^2) = -\sigma^* (1-3(\frac{r}{0.55})^2)$

$\sigma_z = -2\sigma^* (1-2\eta^2) = -2\sigma^* (1-2(\frac{r}{0.55})^2)$

σ_{max} may be at $r=0$ or $r=0.55 \rightarrow \sigma_{max} = 1470 \text{ MPa}$

compressive $\left\{ \begin{aligned} \sigma_{r,0} &= -735(1) \\ \sigma_{\theta,0} &= -735(1) \\ \sigma_{z,0} &= -2 \cdot 735(1) \end{aligned} \right.$

$\sigma_{r,Rf} = -735(1-1) = 0 \text{ MPa}$

$\sigma_{\theta,Rf} = -735(1-3 \cdot 1) = 1470 \text{ MPa}$

$\sigma_{z,Rf} = -2 \cdot 735(1-2 \cdot 1) =$

$= -1470(-1) = 1470 \text{ MPa}$

b) $\sigma_\theta(4\pi) = \sigma^* (1-3\eta^2)$

~~$175 = -735 + 2205\eta^2$~~

$\rightarrow 175 = -735(1-3(\frac{r}{0.55})^2) \rightarrow 0.4127 = (\frac{r}{0.55})^2 \rightarrow r = 0.363 \text{ cm}$

2) $P = 15 \text{ Mpa}$ $\bar{R}_L = 0.55 \text{ cm}$ $\delta_L = 0.05 \text{ cm}$
 $R_i = 0.525 \text{ cm}$ $R_o = 0.575 \text{ cm}$

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a) Thin-walled:

$$\sigma_\theta = \frac{15 \cdot 0.55}{0.05} = 165 \text{ Mpa}$$

$$\sigma_z = \sigma_\theta / 2 = 82.5 \text{ Mpa}$$

$$\sigma_r = -\frac{P}{2} = -7.5 \text{ Mpa}$$

b) Thick-walled @ $r = 0.525 \text{ cm}$

$$\text{same} \left(\left(\frac{R_o}{R_i} \right) = 1.0952 \right)$$

$$\sigma_r(0.525) = -15 \frac{(1.095)^2 - 1}{(1.095)^2 - 1} = -15 \text{ Mpa}$$

$$\sigma_\theta(0.525) = 15 \frac{(1.095)^2 + 1}{(1.095)^2 - 1} = 15 \frac{2.199}{0.199} = 165.75 \text{ Mpa}$$

$$\sigma_z(0.525) = 15 \frac{1}{(1.095)^2 - 1} = 75.37 \text{ Mpa}$$

$$3) R_f = 0.48 \text{ cm}, \quad \delta_{gap} = 0.008 \text{ cm}, \quad T_{co} = 550 \text{ K}, \quad \delta_{clad} = 0.08 \text{ cm}$$

$$k_f = 0.03 \frac{\text{W}}{\text{cmK}}, \quad k_{gap} = 0.003 \frac{\text{W}}{\text{cmK}}, \quad k_{clad} = 0.15 \frac{\text{W}}{\text{cmK}}$$

$$\frac{16}{10} \text{ LHR} = 250 \frac{\text{W}}{\text{cm}}, \quad \alpha_{clad} = 10 \text{E-}4 \text{ K}^{-1}, \quad \alpha_f = 14 \text{E-}6 \text{ K}^{-1}$$

$$T_{ref} = 300 \text{ K}$$

$$R_{ic} = 0.488 \text{ cm}, \quad R_{oc} = \cancel{0.488} 0.548 \text{ cm} \rightarrow \bar{R}_c = 0.528 \text{ cm}$$

$$T_{ic} = \frac{\text{LHR} \delta_{clad}}{2\pi R_f k_{clad}} + T_{oc} = \frac{250 \cdot 0.08}{2\pi(0.48) \cdot 0.15} + 550 = 594.2 \text{ K}$$

$$T_s = \frac{\text{LHR} \delta_{gap}}{2\pi R_f k_{gap}} + T_{ic} = \frac{250 \cdot 0.008}{2\pi(0.48) \cdot 0.003} + 594.2 = 815.2 \text{ K}$$

$$T_o = \frac{\text{LHR}}{4\pi k_f} + T_s = \frac{250}{4\pi \cdot 0.03} + 815.2 = 1478.35 \text{ K}$$

$$\begin{aligned} \rightarrow \bar{T}_c &= \frac{550 + 594.2}{2} = 572.1 \text{ K} \\ \bar{T}_f &= \frac{815.2 + 1478.4}{2} = 1146.8 \text{ K} \end{aligned}$$

→ poorish assumption
(should be integral avg)
- yes, you're right

$$\begin{aligned} \rightarrow \Delta \delta_{gap} &= 0.528 \cdot 10 \text{E-}4 (572.1 - 300) - 0.48 \cdot 14 \text{E-}6 (1146.8 - 300) \\ &= 5.28 \text{E-}4 (272.1) - 4.72 \text{E-}4 (846.8) \\ &= 0.00144 - 0.00569 \end{aligned}$$

$$\Delta \delta_{gap} = -0.00425 \text{ cm}$$

4) $\Delta T = 50K$, $\alpha_c = 15E-6$, $E = 100GPa$, $\nu = 0.34$

$b_c = 0.06cm$, $R_i = 0.55cm$, $R_o = 0.61cm$ ✓

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$$\sigma_\theta = \frac{1}{2}(50) \frac{15E-6 \cdot 100}{(1-0.34)} \left(1 - 2\left(\frac{0.55}{0.06}\right)^2 \left(1 - 1\right)\right)$$

$$= \cancel{54.8 \text{ MPa}}$$

$$\sigma_r = \frac{1}{2}(50) \left(\frac{15E-6 \cdot 100}{1-0.34}\right) \left(1 - 2\left(\frac{0.55}{0.06}\right)^2 \left(\frac{0.55}{0.06}\right)\right)$$

$$\sigma_r = \frac{1}{2}(50) \left(\frac{15E-6 \cdot 100}{1-0.34}\right) \left(\frac{0.55}{0.55} - 1\right) (\dots)$$

$$= 0 \text{ MPa}$$

goes to 1

$$\sigma_\theta = \frac{1}{2}(50) \left(\frac{15E-6 \cdot 100}{1-0.34}\right) \left(1 - 2\frac{0.55}{0.06}(0)\right)$$

$$\sigma_\theta = \cancel{25.4} \quad 0.0568 \text{ GPa} = 56.8 \text{ MPa}$$

$$5) \quad \dot{F} = 6E13 \frac{f}{cc}, \quad \Delta p_o = 0.015, \quad B_o = \frac{5}{950} = 0.0053 \text{ FIMA}$$

$$p(uo_2) = 10.97 \frac{g}{cc}, \quad t = 10 \cdot 24 \cdot 60 \cdot 60 = 8.64E5 s$$

$$\rightarrow \epsilon_D = 0.015 \left(\exp \left(\frac{B \ln(0.01)}{0.0053} \right) - 1 \right)$$

$$\beta = \frac{\dot{F} \cdot t}{Nu}$$

$$Nu = \frac{10.97}{270} \cdot 4.022E23$$

$$\beta = \frac{6E13 \cdot 8.64E5}{2.747E22}$$

$$= 0.00212 \text{ FIMA}$$

$$\frac{\frac{g}{mol} \cdot \frac{\#}{mol}}{\frac{g}{cc}} \cdot \frac{\frac{g}{cc} \cdot \frac{\#}{mol}}{\frac{g}{mol}}$$

$$\rightarrow \epsilon_D = 0.015 \left(\exp \left(\frac{0.00212 \ln(0.01)}{0.0053} \right) - 1 \right)$$

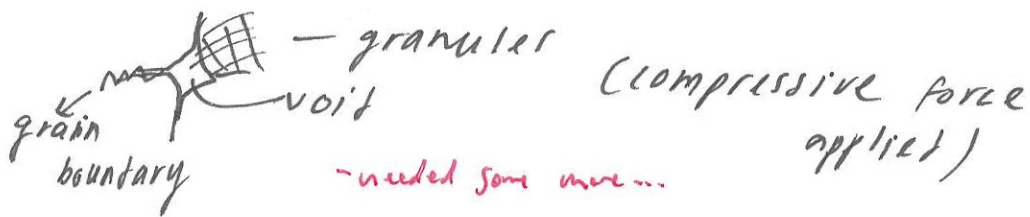
$$\epsilon_D = -0.6124$$

6) Elasticity is a measure of a change in length and its ability to come back to original form, while plasticity represents a permanent change in a material structure upon an applied load.

7) strain hardening is the embrittlement of a material given a continuous strain. Hardening occurs because of dislocation buildup in slip-planes and other imperfections, increasing ~~dislocation~~ energies - kind of... migration

8) Fuel performance codes must calc. ~~heat conduction~~ heat conduction Eqn. in time with other changes in mat. props. due to rad. damage (i'm not sure about 3 specific things) - some of...

9) Grains form when the boundaries of individual granules are formed into one porous mass.



10) Microstructure based modeling involves sort of ab-initio modeling, as it does not employ the use of empirical models. This is beneficial b/c it allows for a more reliable result - kind of - but needed more realistic

11) the microstructure is a gauge on a materials smaller structure (grain orientation, texture, and size?). Annealing is a processing technique that allows for

11) imperfections to migrate to surface and for grains to grow. This allows for a ~~softer~~ more ductile material.
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12) Diffusion occurs more quickly along the grain boundary b/c ~~that~~ the migrating species prefer a larger gap to travel through (energetically)
4/4

13) The driving force of grain growth is the lowering of system energy with less grain boundaries. This can be impeded by other defects.
5/6 - need more specification

14) Fuel densifies b/c of the sintering process. When the fuel is heated and under stress in the reactor, the ~~voids~~ at grain corners close and shrink the structure.
4/6 - need more

